

Team Project

A payload of mass 3000 kg is to be raised from its circular, low earth orbit (LEO) at 300-km altitude to a circular, geosynchronous, equatorial orbit (GEO) using a two-impulse Hohmann transfer. $\mu_E = 398600 \text{ km}^3/\text{s}^2$, $R_E = 6378 \text{ km}$.

(a) Calculate the $(\Delta v)_{\text{reqd}}$ for the first impulse of the Hohmann transfer.

The thrust chamber of an upper stage rocket, using RP-1 and LOX, is to be designed for providing this first impulse. It will use a chamber pressure of 45 atm and area ratio of 100. The payload acceleration is not to exceed $5g_e$ during the application of the impulse. The structural coefficient of the upper stage rocket is 0.065.

(b) Determine the optimum propellant mixture ratio.

(c) Find the mass of the propellant required.

(d) Find the burn time.

(e) Find the thrust of the rocket at impulse initiation.

(f) Determine the throat and exit dimensions of the nozzle.

(g) Determine the volume flow rates of the fuel and the oxidizer.

(h) Find the volume of the fuel and oxidizer tanks. Assume ullage volume to be 5% of the tank volume for both the fuel and the oxidizer tanks.

Both the fuel stream and the oxidizer stream experience a pressure drop of 550 kPa across the injector. The discharge coefficient of the injector orifices is 0.75.

(i) Find the overall injection areas of the fuel stream and the oxidizer stream.

(j) Find the injection velocities of the fuel jet and the oxidizer jet into the combustor.

(k) Given that the injection angle of the oxidizer jet is 20° to the thrust chamber axis, find the injection angle of the fuel jet.

(l) Find the total pump power needed to deliver the fuel and oxidizer to the thrust chamber. Both the fuel and oxidizer tanks are maintained at a pressure of 1 atm.